Using SQLite in a RAG Application

SQLite is a lightweight relational database that does not require a separate database server. The entire database can be stored in a single local file, which makes SQLite convenient for small offline RAG projects.

A RAG application can use a documents table to store information about each source document. Typical fields include the file name, the file path, and a hash of the file contents. The hash can help the application determine whether the same document has already been processed.

A chunks table can store each document passage together with its page number, chunk number, original text, and embedding vector. Because one document can contain many chunks, the relationship between documents and chunks is usually one-to-many.

For a small project, embedding vectors can be stored in SQLite as JSON text. When a query arrives, the application can load the vectors into Python and calculate cosine similarity there. This approach is simple and easy to understand when the document collection is small.

As the knowledge base becomes much larger, comparing a query with every stored vector may become expensive. Large systems often use specialized vector indexes or vector databases. For a small educational RAG project with only a limited number of documents and chunks, SQLite is usually sufficient.